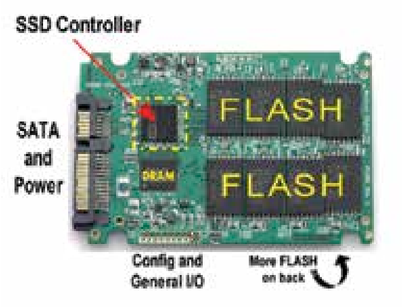

Firmware

The controller is basically a programmable microchip and it is the firmware which tells it how to do it's job. This way whenever newer methods are invented, your old SSD can be updated to follow them. Though you can only use firmware that your SSD manufacturer releases and not the stock firmware that the controller's manufacturer releases. This is because there are customisations performed by each manufacturer. There have been certain open source firmware released by third parties that promise better performance, but using them will void your SSD's warranty.

So how good are SSDs?

We know that they are better in a lot of aspects when compared to hard drives so let's take a look at all the parameters where they excel.

1. Speed

SSDs are much faster than HDDs since access times are way lower than HDDs and because of the fact that any section of the SSD can be read without going through it's adjacent cells. Hard drives have to seek to a location in order to access it, SSDs don't have this problem.

2. Boot-time

Due to the faster transfer speeds and the lack of a spin-up period the boot time when an OS is installed on the SSD is roughly under 30 seconds even after significant wear and tear.

3. Durability

Since SSDs don't have any moving parts at all and that they are lighter than HDDs, they are much more durable and shock resistant. Most SSDs weigh approximately around a 100 gms, even with metal casings they don't weigh more than 300 gms.

4. Noise

Since there are no moving parts, there is no noise at all.

5. Power consumption

At the heart of a HDD lies a very powerful motor and motors consume a lot of electricity. Then there is the read / write head which uses very powerful electromagnets, again something that consumes a lot of power. The SSD on